

2) a) Slip:-

It is the difference between synchronous speed and rotor speed, expressed as a fraction of synchronous speed.

$$s = \frac{N_s - N_r}{N_s} \quad \therefore N_s = \text{synchronous speed} \\ N_r = \text{rotor speed}$$

b) Synchronous Speed:-

It is the speed of the rotating magnetic field produced in the stator.

$$N_s = \frac{120f}{P} \quad \therefore f = \text{supply frequency} \\ P = \text{number of poles}$$

c) Rotor Frequency:-

It is the frequency of the induced EMF in the rotor.

$$f_r = sf \quad \therefore s = \text{slip} \\ f = \text{stator supply frequency}$$

Relation between them:-

- $s = \frac{N_s - N_r}{N_s}$

- $f_r = sf$

This means:-

- If slip = 0 $\rightarrow$ rotor frequency = 0 (no induction).
- If slip increases $\rightarrow$ rotor frequency increases.

3) $f = 50\text{Hz}$
 $P = 4$

$N_r = 1440\text{rpm}$

a) $N_s = \frac{120f}{P} = \frac{120 \times 50}{4}$
 $= 1500\text{rpm}$